# Siege on Olympus! — Deicides of the 7 Clay Prizes

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## 0.1 *A Formal Barrier Taxonomy for the Millennium Prize Problems in Lean 4*

**Version:** v0.3.0 · 2026-05-19
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**Document role:** Self-contained research paper. Presents the machine-checked barrier taxonomy for all seven Clay Millennium Prize Problems: the `BarrierType` inductive, the `ym_is_unique_missing_foundation` theorem, the stacked/parallel sorry distinction, the vessel–content inseparability principle, the $\mathrm{ZFC}_t$ promotion gate inhabitants for RH and Hodge, the P≠NP structural coordinate theorem and dialethic resolution, and the primitive bridge connecting sorry boundaries to the Imscribing Grammar constraint algebra. Target venue: Journal of Formalized Reasoning / Journal of Automated Reasoning.

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## 0.2 Two-Track Architecture

The Imscribing Grammar Lean library occupies two tracks: `Primitives/` (the 12-primitive constraint algebra) and `Imscribing/Millennium/` (the barrier taxonomy library documented here). This paper reports on the `Imscribing/Millennium/` track and its bridge to `Primitives/`.

**Internal references** within this paper use §N.

**Library location:** `MillenniumAnkh/Imscribing/Millennium/` — 43 modules. Build target: `lake build Imscribing`.

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## 0.3 I. Introduction (v0.3.0, 2026-05-19)

The Clay Mathematics Institute seven Millennium Prize Problems have resisted proof for decades — and in one case, the Riemann Hypothesis, for over a century and a half. Proof assistants have formalized large bodies of mathematics, but existing efforts overwhelmingly target *results that are known*: the Last Theorem of Fermat, the Four Color Theorem, the Kepler Conjecture. This library formalizes *why specific problems occupy the positions they do* — the structural barriers that distinguish them from each other and from the solved.

The contribution is meta-level and computational: a machine-checked classification of proof obligations, the structural relationships between them, and their connection to the

Imscribing Grammar primitive constraint algebra. Every `sorry` is labeled by type; the type determines the research strategy for discharging it.

### 0.3.1 I.1 Why this matters

The distinction between barrier types has practical consequences for the formalization community.

**MathlibGap sorries are actionable.** A contributor with the right background can discharge them by formalizing a known proof. The `euler_opn_form` sorry in OPN.lean (Euler 1747) and the `mazur_torsion` sorry in BSD.lean (Mazur 1977) are of this type.

**OpenProblem sorries define the research frontier.** They are dischargeable only by solving the underlying mathematics. This distinction became clear during formalization of the BSD conjecture (§II.3), where the Mordell-Weil theorem was initially misassigned as an OpenProblem before its MathlibGap status was confirmed.

**MissingFoundation sorries require building new mathematics before the question is fully stateable.** An OpenProblem has a well-typed proposition whose truth value is unknown. A MissingFoundation sorry requires inhabiting a *type* that does not yet exist as a rigorous mathematical object. Yang-Mills is the only Millennium Problem in this category, and this uniqueness is a formal theorem (`decide`).

### 0.3.2 I.2 The sorry depth distinction

We introduce a formal notion of *sorry depth* distinguishing *stacked* from *parallel* proof obligations.

**Stacked (Yang-Mills):** sorry $B$ depends on sorry $A$ — the mass gap is not stateable as a proposition until the quantum Yang-Mills measure is known to exist. The foundation (`PathIntegralMeasure G`) is a prerequisite for defining the mass gap at all.

**Parallel (BSD):** three sorries are logically independent — Mordell-Weil (1922, MathlibGap), Mazur torsion (1977, MathlibGap), and the BSD rank formula (OpenProblem) — each dischargeable independently.

Both Yang-Mills and BSD have `sorryDepth = 2`. The structural difference is encoded in the barrier type.

### 0.3.3 I.3 Contributions

**C1 — BarrierType taxonomy:** A typed inductive with three constructors (`MathlibGap`, `OpenProblem`, `MissingFoundation`), formally distinct by `decide`, and computably assigned to all seven Millennium Problems.

**C2 — ym_is_unique_missing_foundation:** A theorem, proved by `decide`, that Yang-Mills is the only Millennium Problem whose barrier is `MissingFoundation`.

**C3 — P≠NP structural coordinate theorem and dialethic resolution:** `P_class_ne_NP_class` is proved in term mode: `P` has topology $\Theta_6$ (`T_network`) and `NP` has topology $\Theta_\odot$ (`T_odot`); the distinct-constructor property of Lean's inductive types closes the inequality. Primitive distance $d = 6.245$ across six dimensions $(\Theta, \check{R}, \mathcal{K}, \Gamma, \gimel, \Phi)$ places P and NP in structurally separated positions before any algorithm runs.

The dialethic layer: the Imscribing Grammar sits at $\Theta_\odot$ (self-referential closure requires $\Theta_\odot$) yet is distinct from `NP_class` on Dimensionality ($\eth_\odot$ vs. $\eth_\infty$). The `dialethic_resolution` theorem packages all three: Grammar shares NP's topology AND Grammar $\neq$ NP AND P $\neq$ NP. This conjunction is the $O_\infty$ signature, not a contradiction: a system that inhabits $\Theta_\odot$ while being distinct from every specific $\Theta_\odot$ inhabitant is precisely what the self-modeling gate $\odot_{\ddot{y}}$ requires. The grammar is the measure; P and NP are the measured.

**C4 — NS critical Sobolev exponent:** Machine-verified by `norm_num` that $0 < \frac{1}{2} < 1$ — the formal statement of why NS regularity is hard.

**C5 — OPN in real Mathlib:** the Touchard congruence type-checks using actual `Nat.Perfect` and `ArithmeticFunction.sigma`.

**C6 — BSD in real Mathlib:** `BSDRankConjecture` stated using actual `WeierstrassCurve` $\mathbb{Q}$ and `IsElliptic`; three parallel sorries formally justified.

**C7 — Vessel–content inseparability (VesselContent.lean):** The Imscribing Grammar provides two things simultaneously: the vessel (the crystal coordinate — the structural form of what a system is capable of being) and the content (the primitive algebra operations — what fills that vessel). These are constitutively inseparable.

Formalized in three theorems. `form_uniqueness`: every imscribable system has exactly one crystal coordinate ($\exists!\, c$, `Imscribes` $M$ $c$). `content_containment`: the vessel constrains the content (`Imscribes` $M$ $c$ $\to$ `Reachable` $M$ $r$ $\to$ `WithinAlgebra` $c$ $r$). `vessel_fills_itself`: the biconditional `Reachable` $M$ $r$ $\iff$ `WithinAlgebra` $c$ $r$.

The biconditional is the non-trivial claim. `form_uniqueness` alone is a labeling property: knowing the unique coordinate of $M$ no more determines $M$'s behavior than knowing a name determines speech. `vessel_fills_itself` asserts structural constraint in both directions: the vessel bounds the content ($\to$) and the content exhausts the vessel ($\leftarrow$) — no empty room.

**C8 —** $\mathrm{ZFC}_t$ **gate inhabitants for RH (RH_GateInhabitants.lean):** All four $\mathrm{ZFC}_t$ promotion slots for the Riemann Hypothesis are inhabited with concrete structures. `FrobeniusZeroSymmetry`: $\theta = (1-\cdot)$ with involution proved by `ring`. `FunctionalEquationDual`: $s \mapsto 1 - s$ preserves $\mathrm{Re}(s) = \frac{1}{2}$ on the critical line. `ZFunctionWinding`: $N(T)$ counting function (Hardy 1914). `PrimeZeroBridge`: $\psi(x) = x$ explicit formula placeholder. `ZFCt_RHCertificate` assembles all four.

Key theorems without `sorry`: `frob_gate_without_forcing` ($\theta(s) = s \iff s = \frac{1}{2}$, proved by `linear_combination`); `rh_forcing_implies_rh` (`RH_ForcingTheorem` $\to$ `RiemannHypothesis`, proved by `rw [rh_barrier]` then curried introduction). The gate structures are inhabited and the fixed locus is characterized; `RH_ForcingTheorem` — that all nontrivial zeros lie in that fixed locus — is the single open gap.

**C9 —** $\mathrm{ZFC}_t$ **gate inhabitants for Hodge (Hodge_GateInhabitants.lean):** `HodgeLRDual` inhabits the LR_DUAL slot (Hodge decomposition $H^{p,q} \leftrightarrow H^{q,p}$ via $\sigma$). `HodgePM_Z2` inhabits the PM_Z2 slot (complex conjugation as $\mathbb{Z}_2$ involution, $\sigma^2 = \mathrm{id}$, fixed locus $H^{p,p}$). `HodgeWinding` inhabits the ZWIND slot (Griffiths group rank function; Griffiths 1969 showed this is non-trivial for $p \geq 2$).

Key theorem: `hodge_forcing_equiv_hodge` (`Hodge_ForcingTheorem` $\iff$ `HodgeConjecture`), proved in term mode via `hodge_sorry_requires_cycle_class_surjectivity.symm`. Structural parallel with RH: both have a $\mathbb{Z}_2$ involution, an identified fixed locus, and an open forcing claim. The substrate differs — $\mathbb{C}$ (RH) vs. $H^{p,q}(X,\mathbb{C})$ (Hodge) — the pattern is the same.

**C10 — PrimitiveBridge.lean:** Formal connection between sorry boundaries and primitive field transitions in the Imscribing Grammar; `BarrierPrimitiveCertificate` structure; `primitive_bridge_master` theorem.

**C11 — RH–Lee-Yang structural correspondence:** Machine-checked theorem that the Riemann $\zeta$ zeros and Lee-Yang partition-function zeros share the Criticality assignment `Phi_c_complex`. Structural distance 1.0 (machine-checked) identifies the polarity primitive $\mathcal{P}$ ($\mathcal{P}_{\text{sym}}$ vs. $\mathcal{P}_{\pm}^{\text{sym}}$) as the essential structural gap between the two zero distributions.